

Project Review 1 Report

Project Title: Adaptive Routing in Energy-Harvesting Wireless Sensor Networks
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Repository: `https://github.com/santhoshvellore7119-web/WSN_proj``

1. Problem Statement & Objectives

In conventional Wireless Sensor Networks (WSNs), sensor nodes are powered by non-rechargeable electrochemical batteries, meaning residual energy monotonically decreases. Standard routing algorithms (e.g. Dijkstra, LEACH, static bottleneck DP) assume static battery snapshots, routing traffic away from any node with depleted reserves.

When nodes harvest ambient energy (solar irradiance, thermal gradients, or ambient RF), battery levels fluctuate dynamically. A node with low residual energy at the start of a transmission round can recharge just-in-time during multi-hop transit. Blindness to incoming energy leads to:

- False Rejection of Recharging Nodes:** Standard shortest paths avoid nodes that would be fully charged upon packet arrival.
- Burnout of Non-Harvesting Nodes:** Stagnant, shaded nodes carrying high initial charges are overused until premature depletion.

Core Project Objectives:

- Algorithmic Contribution:** Formulate and implement **Time-Augmented Maximin Dynamic Programming** ($\text{dp}[v][h][t]$) to maximize bottleneck route capacity on a time-expanded acyclic state graph.
- Fast Fault Recovery:** Implement **Disjoint-Set Union (Union-Find)** with path compression and union-by-rank for local detour rerouting ($O(|E|\alpha(V)) + O(\deg(u)\alpha(V))$).
- Literature Baselines:** Benchmark against published EH-WSN protocols (**EH-LEACH** and predictive energy-weighted shortest path).
- Empirical Rigor:** Evaluate under calibrated 24-hour solar irradiance traces (derived from NREL NSRDB seasonal datasets), spatial heterogeneity sweeps ($p_{\text{shadow}} \in [0.0, 1.0]$), and multi-seed statistical significance tests (paired Wilcoxon signed-rank and Student's t-test).
- Full-Stack Engineering:** Provide 2 unified interfaces (CLI research tools and FastAPI + React web application) with full-stack CI and Docker containerization.

2. DSA Architecture & Modules

The simulation engine is structured around foundational Data Structures and Algorithms:

```
wsn-energy-routing/
├── src/
│   ├── network.py           # Node & Graph adjacency list representations
│   ├── energy_model.py      # LEACH 1st-order radio model (d^2 free-space / d^4 multipath)
│   ├── harvesting_model.py   # Solar, Stochastic Poisson, RealTrace (NREL), Shadowed & RF profiles
│   ├── clustering.py         # LEACH and EH-LEACH clustering with min-heap tiebreaking
│   ├── routing.py           # Dijkstra, A*, Energy-Aware Dijkstra, Union-Find live detour engine
│   ├── dp_lifetime.py       # 2D Classical Maximin DP and 3D Time-Augmented DP (dp[v][h][t])
│   ├── simulator.py         # 5-phase discrete event simulation engine
│   └── visualize.py         # Matplotlib plotting routines and heatmap generators
├── backend/                 # FastAPI REST API with async background tasks and SQLite models
├── frontend/                # React dashboard with interactive network SVG and Recharts graphs
└── tests/                   # Pytest suite (50 unit tests covering core + baselines + API)
```

```

■■■ Dockerfile.backend      # Python 3.12 container definition
■■■ Dockerfile.frontend    # Node 18 / Nginx container definition
■■■ docker-compose.yml     # Single-command full-stack deployment

```

Key Complexity & State Formulations:

- **Graph Topology** (`network.py`): Adjacency list storing distance-weighted edges.
- **Time-Augmented DP** (`dp_lifetime.py`): $\text{D Table } dp[v][h][t]$ on a time-expanded DAG with recurrence:

$$dp[v][h][t] = \max_{u \in \text{nbr}(v)} \min \left(dp[u][h-1][t-\delta], \lambda, E_{\text{proj}}(v, t_{\text{curr}} + t) \right)$$
- **Time Complexity**: $O(|E| \cdot H \cdot T)$
- **Space Complexity**: $O(|V| \cdot H \cdot T)$
- **Union-Find Detour Recovery** (`routing.py`): $O(|E| \cdot \alpha(V))$ setup, $O(\deg(u) \cdot \alpha(V))$ local detour repair, achieving a $6.1\times$ latency reduction over full DP recomputation.

3. Experimental Validation & Results

3.1 Unit Testing

All **50 unit tests** pass with `pytest` in $< 3.5\text{s}$, verifying:

- Radio energy calculations and crossover threshold $d_0 = \sqrt{E_{\text{fs}}/E_{\text{mp}}}$ $\approx 87.7\text{m}$.
- Deterministic 5-node adversarial counterexample isolating the lookahead mechanism.
- Classical and Time-Augmented DP table filling, battery clamping, and backpointer reconstruction.
- Disjoint-Set Union connectivity, rank optimization, and local detour splicing.
- Literature baselines: EH-LEACH election and RealTrace solar sampling.
- FastAPI backend routes (`/health`, `/simulate`, `/benchmark`, `/runs`, `/experiments/scalability`, `/experiments/heterogeneity`, `/simulate/{job_id}/csv`) via `TestClient`.

3.2 Canonical Benchmark (50 Nodes, 350 Rounds, $R_{\text{tx}} = 35.0\text{m}$, Seed 42)

Configuration	First Node Death (FND)	Half Nodes Dead (HND)	Alive (Round 350)	Residual Energy (J)
1. Baseline (No Harvesting)	Round 92	Round 108	0 / 50	0.0000 J
2. Solar (Unaware LEACH + Dijkstra)	Round 182	Round 205	0 / 50	0.0000 J
3. Solar (Adaptive Time-DP)	Round 151	Round 194	1 / 50	0.0276 J
4. Shadowed (Unaware Dijkstra)	Round 112	> 350	27 / 50	0.6630 J
5. Shadowed (Adaptive Time-DP)	Round 107	> 350	27 / 50	0.5930 J
6. Stochastic Poisson (Unaware LEACH)	Round 302	> 350	36 / 50	0.0963 J
7. Stochastic Poisson (Adaptive Time-DP)	Round 302	> 350	26 / 50	0.0479 J

3.3 Harvesting Heterogeneity Sweep ($p_{\text{shadow}} \in [0.0, 1.0]$)

- **Empirical Findings & Trade-Off Analysis**:
 - In low-to-moderate spatial occlusion ($p_{\text{shadow}} \leq 0.8$), multi-hop relaying introduces an unavoidable reception dissipation penalty ($E_{\text{rx}} = k \cdot E_{\text{elec}}$) on intermediate relays. Direct-to-sink transmission minimizes aggregate reception energy when nodes can reach the base station.
 - In extreme occlusion ($p_{\text{shadow}} = 1.0$), where all active relays suffer heavy solar deprivation, Time-Augmented DP's lookahead routing actively protects vulnerable forwarders, extending network operational lifetime by **+6 rounds** (FND 102 vs 96) and preserving greater residual energy (+1.2% gain, 0.0866 J vs 0.0855 J).
- **Stochastic Regime Regression** (36 vs 26 Alive Nodes): In stochastic Poisson harvesting ($\lambda = 2.0$, $q = 0.00015\text{ J}$), discrete arrival bursts cause realized harvest to periodically fall short of expected projections. Relays

accepting forwarding tasks under optimistic lookahead projections deplete prematurely, leaving 26 surviving nodes compared to 36 under static shortest-path Dijkstra.

3.4 Multi-Seed Statistical Validation (\$N = 30\$ Seeds, 350 Rounds)

Evaluated across 30 independent topologies (seeds 42, 7, 123, ..., 2026) to eliminate random deployment bias:

- **Solar Regime:** $\Delta E = -0.0751 \text{ J}$ (95% CI: $[-0.0829, -0.0673]$, paired $t = -18.89$, $p = 7.66 \times 10^{-18}$, Wilcoxon $W = 0$, $p = 1.86 \times 10^{-9}$). Unaware LEACH preserves more aggregate energy due to avoiding intermediate relay reception dissipation (E_{rx}).
- **Shadowed Solar:** $\Delta E = -0.0380 \text{ J}$ (95% CI: $[-0.0453, -0.0306]$, $p = 5.15 \times 10^{-11}$).
- **Stochastic Regime:** $\Delta E = -0.0709 \text{ J}$ (95% CI: $[-0.1321, -0.0097]$, $p = 0.0308$).
- **Conclusion:** Single-path maximin bottleneck optimality (Theorem 1) provides worst-case path resilience, but macroscopic network energy depends on the multi-hop relay dissipation trade-off.

3.5 Scalability Benchmark (\$N = 50\$ to 500 Nodes)

Empirical latency scaling measurements confirm theoretical asymptotic complexity:

- Dijkstra: 0.15 ms ($N=50$) to 37.2 ms ($N=500$)
- Classical DP: 4.8 ms ($N=50$) to 168.7 ms ($N=500$)
- Time-Augmented DP ($T=10$, $H=5$): 10.2 ms ($N=50$) to 217.2 ms ($N=500$)
- DSU Local Detour: 1.7 ms ($N=50$) to 23.5 ms ($N=500$) — achieving a consistent $\sim 6.1x$ speedup over full Time-DP recalculation.

4. Modeling Assumptions & Threats to Validity

To ensure scientific honesty and rigor, the boundaries and physical trade-offs of the simulation are explicitly identified:

1. **Radio Model & Relay Reception Overhead:** First-order radio equations (d^2 / d^4) impose $E_{\text{rx}} = k \cdot E_{\text{elec}}$ on every intermediate relay. Multi-hop routing through harvesting bridges is advantageous only when the energy harvested by the relay or the distance savings from multipath (d^4 to d^2) outweigh the cumulative E_{rx} reception cost.
2. **Timescale Discrepancy:** Ambient solar harvesting operates over diurnal cycles (hours), whereas packet propagation across hops occurs on millisecond scales. Lookahead routing assumes discrete scheduling epochs or delay-tolerant buffering between rounds.
3. **MAC Layer:** Assumes an idealized collision-free TDMA schedule within clusters and non-interfering orthogonal CDMA channels across clusters.
4. **Mobility:** Sensor nodes and the sink are statically deployed.
5. **Time Synchronization:** Assumes loose coarse-grained round synchronization sufficient for discrete lookahead intervals Δt .

5. Summary & Next Steps for Review 2

1. **Current Status:** Core algorithms, literature baselines, multi-seed statistical suites, empirical scalability tests, REST API backend, React frontend, CI workflows, and Docker orchestration are fully implemented and verified.
2. **Plan for Review 2:**
 - Multi-packet burst transmission and queueing delay modeling.
 - Dynamic channel fading and shadow margin modeling.